Engineering Mathematics III

ENGE702

Lecture 6: Second Order Homogeneous ODEs

Second order ODEs

In this lecture we will look at second order ODEs.

Remember that a **second order ODE** is an ordinary differential equation where the highest derivative is a second derivative.

A second order ODE is **linear** if it can be written in the form:

$$y'' + p(x)y' + q(x)y = r(x).$$

A second order linear ODE is **homogeneous** if $r(x) = 0$, ie it can be written in the form

$$y'' + p(x)y' + q(x)y = 0.$$

Second order ODEs

One fundamental fact about solutions to homogeneous linear ODEs is that linear combinations of solutions are also solutions.

That means that the sum of two solutions is also a solution, and a scalar multiple of a solution is also a solution.

Example: The functions $y = \sin(x)$ and $y = \cos(x)$ are solutions to the differential equation

$$y'' + y = 0$$

(check this).

$$y = a\sin(x) + b\cos(x)$$

is also a solution to this differential equation (check this).

This is called the **superposition principle**.

Note that this does not work for non-homgeneous ODEs in general

Second order ODEs

A **general solution** for a a second order homogeneous linear ODE is of the form

$$y = c_1 y_1 + c_2 y_2$$

where y_1 and y_2 are linearly independent solutions of the ODE, and c_1 and c_2 are arbitrary constants.

Showing functions are linearly independent can be a bit complicated in general, but when there are only two (as here for a second order ODE), it just means they are not scalar multiples of each other.

Second order ODEs

Just like with first order ODEs, we also have **initial value problems** for second order ODEs. Now instead of having one initial value, we have two. These are usually given as the value of the function at a point, and the value of a the derivative at that point. From there we can work out values for c_1 and c_2.

Solving an initial value problem involves three steps:

1. Finding two linearly independent solutions. This is usually the tricky bit - we will see techniques of how to do this next lecture.

2. Finding the general solution. This is easy - just put your solutions from step 1 in the form $y = c_1 y_1 + c_2 y_2$.

3. Using the initial conditions to find the values of c_1 and c_2.

Second order ODEs

Example: Solve the initial value problem

$$y'' + y = 0, \qquad y(0) = 1, \quad y'(0) = 0$$

Exercises

- Check the solutions in the first example (superposition principle).
- Solve the initial value problem

$$y'' - 3y' + 2y = 0, \qquad y(0) = 0, \quad y'(0) = 1$$

given $y_1 = e^x$ and $y_2 = e^{2x}$ are both solutions.

Text Book References

Chapter 2.1